PCES BROKER (WEB API) DOCUMENTATION

PPH Application Program Interface

Abstract

Application program interface documentation describing the interface for integration of third party platforms to PCES BROKER

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Change Control

Version	Date	Revised by	Changes
1.0	12.07.2021	Kwabena	Initial version
1.1	12.07.2021	Kwabena	Added Endpoints for GAG Integrations
1.2	27.07.2021	Faisal	Remove booking in Finacle for cash code API for GAG integration
1.3	10.09.2021	Efua	Update Bank to Wallet request/response
1.4	11.01.2022	Efua	Withdrawal by code redemption/reversal and bank to wallet status

Reference Documents

Version	Date	Owner	Document name	Location

1.1. Introduction

This document describes the application program interface of PCES BROKER (Web API) and provides information for third parties integrating with the PCES BROKER.

Methods	Description
Generate ATM Code	Generates a cardless withdrawal code for cash withdrawal on ATM
MPESA Bank to Wallet	Send money from bank account to a MPESA mobile money wallet
Send SMS	Send an SMS to a phone number

Pre-requisite to access the PCES Broker Web API services:

- IP Whitelisting
- VPN connection
- UserID (will be provided by PCES)
- Password (will be provided by PCES)

1.2. Access Details

IP Whitelisting	Contact Bank IT
URL(live)	Contact Bank IT
UserID	Contact Bank IT
Password	Contact Bank IT
URL(Test)	Contact Bank IT
Port(Test)	Contact Bank IT

1.3. Authentication

All endpoints are secured with *OAuth 2.0* and so before accessing any secured endpoints, you will require a bearer token.

The endpoint path to get an access token is: **/token** and expected payload configurations are:

Method: **Post**

"headers": {"content-type": "application/x-www-form-urlencoded"}

Parameters:

grant_type (value should be **password**)

username (your username)

password (your password)

The response for this request is:

{

"access_token": accesstokenvalue

"token_type": "bearer",

"expires_in": 86399

}

Once a token is obtained, it can be used in subsequent requests by parsing it to the Authorization Key in your header.

"headers" : **{"authorization":"token_type accesstokenvalue"}**

Note the space between "token_type" and "accesstokenvalue"

1.4. General Response Format

For responses, the PPH return ***HttpResponseMessage*** and use the ***StatusCode*** property of the same to indicate a success or failure of the request. The payload/response value is contained within the Content property of the response.

For successful response, HTTP status code is "**200"** or ***"OK"*** and the opposite is true for a failed response. In case of a failed response, the content property contains the following structure:

{

"Code": integer denoting what kind of failure this is

"Message": Brief information about the failure

"Description": In some cases, this includes detailed information about the error

}

1.5. Response Codes

The code part of responses are the following in the table below.

Code	Description
100	Pending
200	Success

300	AuthenticationError
400	MissingParameter
500	SystemError
600	NotFound
700	ValidationError

1.6. API Methods

This section provides details of the RESTFULL API of PPH. The methods calls are made in the form of standard HTTP request using POST. All endpoints require authentication

2. Withdrawal by code endpoints

2.1.1. getatmcode

Generates a withdrawal code that can be used for a cardless transaction on the ATM. The code can be used to withdraw cash from the ATM without the need of using a card. PPH sends the generated withdrawal code via SMS to the beneficiary phone number. The code is sensitive data, and therefore the method does not return the code in its response but rather only returns a response status if the code generation was successful and SMS sent to beneficiary or not.

Path: ***POST api/gag/getatmcode***

Request Information

Body Parameters

Name	Type	Description
AccountNumber	String	Account number to be debited once withdrawal is made at the ATM
BeneficiaryPhoneNumber	String	Beneficiary phone number in international format (+243XXXX…)
Amount	Decimal	Amount involved in the transaction
Currency	Char (3)	Char 3 ISO currency code
TransactionID	String	Reference transaction ID from source request to be used in reconciliations

Request Formats

application/json, text/json

Sample:


```
{
 "AccountNumber": "sample string 2",
 "BeneficiaryPhoneNumber": "sample string 2",
 "Amount": 3.0,
 “Currency”: “CDF”,
 "TransactionID": "sample string 4"
}
```

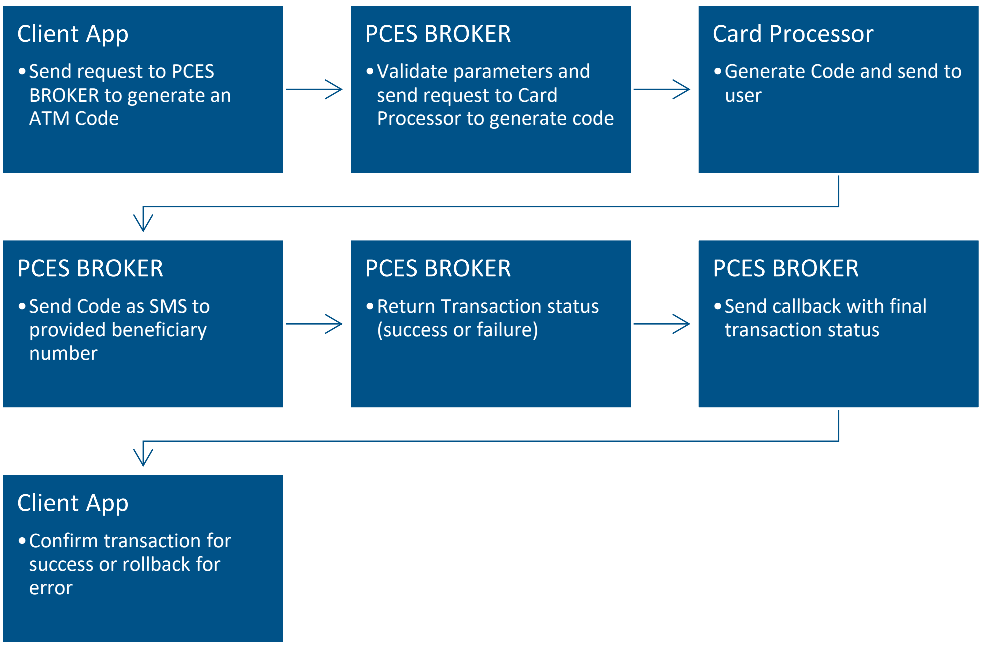

Response Information

application/json, text/json

Sample:


```
{
  OriginTransactionID : “string”,
  TransactionID : “string”
}
```


FLOW



3. ATM Code Redemption and Reversal

3.1. Make Redemption Process Flow

Payment Hub

- Receive notification of used code
- Update code status
- Send information about used code

→

Client App

- Update code status
- Respond with echo of same request to denote success

7.2. makeredemption

This method sends a request with information about a used code to client app.

Request Information

Body Parameters

Name	Type	Description
BeneficiaryPhoneNumber	String	Beneficiary phone number
Currency	String	ISO currency for this transaction
Amount	Decimal	Amount involved in transaction
TransactionID	String	Equity bank transaction ID/reference
OriginatingTransactionID	String	Reference transaction ID from source request to be used in reconciliations
Operation	String	Description of process (usually Redemption)
Resultcode	Decimal	Process code (usually 00)
ResultDescrition	String	Description of process (usually Redemption)

Request Formats

application/json, text/json

Sample:


```
{
"BeneficiaryPhoneNumber": "sample string",
"Currency": "USD",
"Amount": 30,
"TransactionID": "sample string",
"OriginatingTransactionID": "sample string",
"Operation": "Redemption",
"ResultCode": "00",
"ResultDescription": "Redemption"
}
```


Response Information

application/json, text/json

Sample:


```
{
"BeneficiaryPhoneNumber": "sample string",
"Currency": "USD",
"Amount": 30,
"TransactionID": "sample string",
"OriginatingTransactionID": "sample string",
"Operation": "Redemption",
"ResultCode": "00",
"ResultDescription": "Redemption"
}
```


3.3. makereversal

This method sends a request with information about an expired code to client app.

Request Information

Body Parameters

Name	Type	Description
BeneficiaryPhoneNumber	String	Beneficiary phone number in international format
Currency	String	ISO currency for this transaction

Amount	Decimal	Amount involved in transaction
TransactionID	String	Equity bank transaction ID/reference
OriginatingTransactionID	String	Reference transaction ID from source request to be used in reconciliations
Operation	String	Description of process (usually Reversal)
Resultcode	Decimal	Process code (usually 01)
ResultDescrition	String	Description of process (usually Reversal)

Request Formats

application/json, text/json

Sample:


```
{
"BeneficiaryPhoneNumber": "sample string",
"Currency": "USD",
"Amount": 30,
"TransactionID": "sample string",
"OriginatingTransactionID": "sample string",
"Operation": "Reversal",
"ResultCode": "01",
"ResultDescription": "Reversal"
}
```


Response Information

application/json, text/json

Sample:


```
{
"BeneficiaryPhoneNumber": "sample string",
"Currency": "USD",
"Amount": 30,
"TransactionID": "sample string",
"OriginatingTransactionID": "sample string",
"Operation": "Reversal",
"ResultCode": "01",
"ResultDescription": "Reversal"
}
```

4. Check Code Status

Path: *POST api/gag/codestatus*

This method is used to check the status of code.

Name	Type	Description
TransactionID	String	Equity bank transaction ID/reference

Request Formats

application/json, text/json

Sample:

```
{
    "TransactionID": "Sample String"
}
```

Response Information

application/json, text/json

Sample:

```
{
    "TransactionID": "Sample String",
    "Status": "Sample string"
}
```

Name	Type	Description
TransactionID	String	Equity bank transaction ID/reference
Status	String	Current Status of Code (USED, GENERATED, EXPIRED)

5. Bank to Wallet endpoints

5.1.1. Mpesa Bank to Wallet

Transfer funds from bank account to mobile money wallet (mPESA, TigoCash, Africell). PPH debits bank account in the core banking system and credit mobile money wallet. This endpoint is asynchronous and so another request with same transaction ID needs to be made to verify transaction status

Path: *POST api/gag/banktowallet*

Request Information

Body Parameters

Name	Type	Description
BeneficiaryPhoneNumber	String	Phone Number of client in international format (+2XXXXXX...)
Amount	Decimal	Amount involved in transaction
Currency	Char (3)	Char 3 ISO currency code
TransactionID	String	Reference transaction ID from source request to be used in reconciliations
ProviderName	String	Identifier for network provider (Vodacom, Airtel, Africell)

Request Formats

application/json, text/json

Sample:

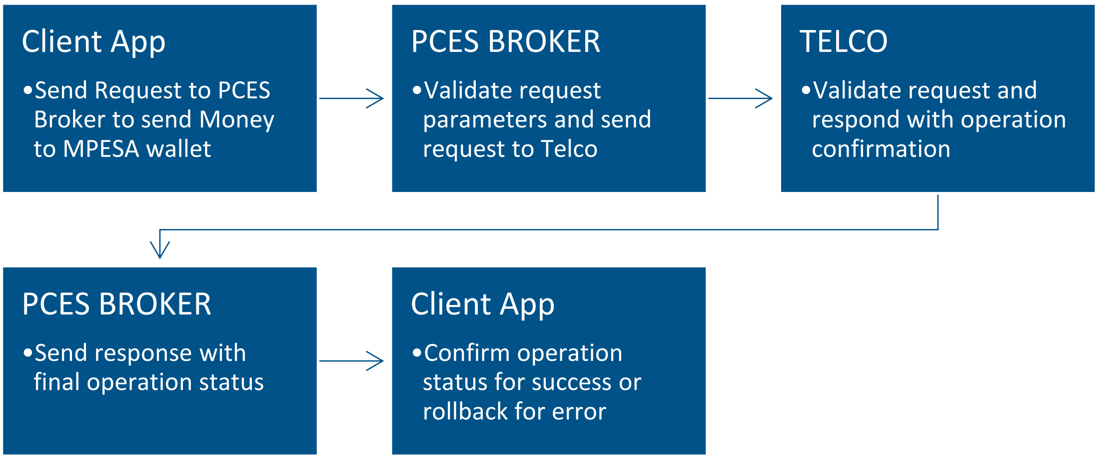

```
{
 "BeneficiaryPhoneNumber": "sample string 2",
  "Amount": 3.0,
  "ProviderName": “sample string 3”,
 "Currency": "CDF",
 "TransactionID": "sample string 4"

}
```

Response Information

application/json, text/json

Sample:

```
{
       OriginTransactionID :"string" ,
      TransactionID : "string"
}
```



6. Check Bank To Wallet Status

Path: *POST api/gag/banktowalletstatus*

This method is used to check the status of bank to wallet transaction.

Name	Type	Description
OriginTransactionID	String	Reference transaction from source request

Request Formats

application/json, text/json

Sample:

```
{
    "OriginTransactionID": "Sample String"
}
```

Response Information

application/json, text/json

Sample:

```
{
    "TransactionID": "Sample String",
    "OriginTransactionID": "Sample String"
    "Status": "OK"
}
```

Name	Type	Description
TransactionID	String	Equity bank transaction ID/reference
Status	String	Status message , OK
`OriginTransactionID`	String	Reference transaction from source request

7. SMS Endpoints

7.1.1. sendsms

This method sends a sms to the provided phone number

Path: *POST api/sendsms*

Request Information

Body Parameters

Name	Type	Description
PhoneNumber	String	Phone number in international format
ProviderName	String	Identifier for network provider (Orange, Airtel etc) In case the number is an international number, specify “International” as provider name
Text	String	Content of SMS message

Request Formats

application/json, text/json

Sample:

```
{
  "PhoneNumber": "+243XXXXXXXXXX",
“ProviderName”: “Orange”,
“Text”:”sample string”
}
```

Response Information

application/json, text/json

Sample:

```
{MessageId:”sample string”}
```

8. Architecture Overview for GAT Implementation